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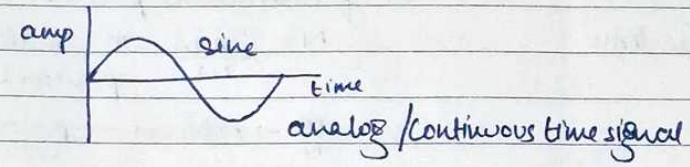
Signals & Systems

UNIT-01.

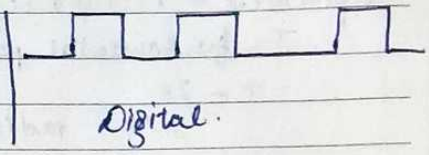
Classification of signals.

↳ Analog. defined for every instants of time, continuous in amp & time. (x(t))

↳ Digital Discrete in time & quantized in amplitude.

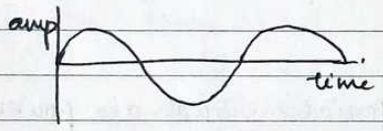


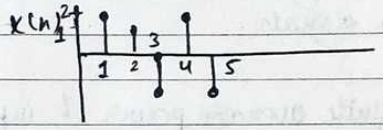
analog / continuous time signal

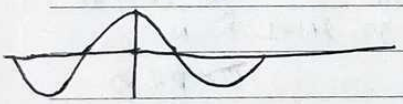
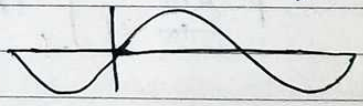


Digital.

Continuous time Signal	Discrete Time signal
• defined for every instant of time.	• defined for discrete instant of time.
• Continuous in amplitude.	• Continuous in amplitude
• Continuous in time	• Discrete in time
• denoted → x(t).	• Denoted → x(n).





Even (Symmetric)	Odd (Anti-symmetric).
CONTINUOUS DOMAIN.	
$x(t) = x(-t)$.	$x(t) = -x(-t)$.
exhibits symmetry w.r.t $t=0$.	exhibits anti-symmetry w.r.t $t=0$
$\frac{x(t)+x(-t)}{2} = x_e(t)$.	$x_o(t) = \frac{x(t)-x(-t)}{2}$.
	
DISCRETE	
$x[n] = x[-n]$.	$x[n] = -x[-n]$.
Signal exhibits symmetry w.r.t $n=0$.	exhibits anti-symmetry w.r.t $n=0$.
$x_e[n] = \frac{x[n]+x[-n]}{2}$.	$x_o[n] = \frac{x[n]-x[-n]}{2}$.

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periodic - Aperiodic Signal.

periodic ⇒ if it repeats again & again over a certain period of time.

aperiodic ⇒ doesn't repeat at a definite interval of time.

CONTINUOUS

periodic = $x(t) = x(t+T)$.
for all t, $T > 0$.

aperiodic - $x(t) \neq x(t+T)$.

T = fundamental periodic time

$T = \frac{2\pi}{\omega}$ → rad/sec.

DISCRETE.

periodic = $x[n] = x[n+N]$.
for all n, $N > 0$.

N = fundamental period.

$N = \frac{2\pi m}{\omega}$ m = smallest positive int.

- $\frac{m}{n}$ → rational → periodic with period $N = m$.
- $\frac{m}{n}$ → irrational → aperiodic.

Energy - power signal.

energy :- finite energy & zero average power. The non periodic signals like exponential signals will have const. energy & so non periodic signals are energy signals.

power :- finite average power & infinite energy. periodic signals are power signals.

CONTINUOUS

energy :- $E = \int_{-\infty}^{+\infty} |x(t)|^2 dt < \infty$.

power :- $P = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} |x(t)|^2 dt$.
$0 < P < \infty$.

DISCRETE

- $E = \sum_{n=-\infty}^{+\infty} |x[n]|^2 < \infty$.
- $P = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |x[n]|^2$.
$0 < P < \infty$.

Deterministic & Random signal.

Deterministic

- exact value at every time is fully predictable.
- no uncertainty.

Random.

- value at any time cannot be predicted exactly, only its probab can be known.
- changes every time it is observed.

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Causal- Non-causal signal.

CONTINUOUS

Causal:- if it is defined for $t \geq 0$.
$x(t) = 0$ for all $t < 0$.
- signal starts at $t = 0$ or later.
- No value exists before time zero.

anti causal:- defined for $t < 0$ or for both. $t < 0$ & $t \geq 0$.
$x(t) \neq 0$ for $t < 0$

DISCRETE.

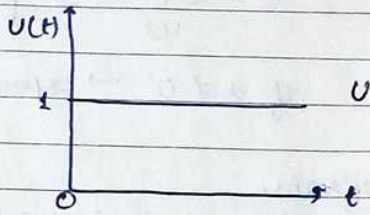
Causal = $n \geq 0$ defined.
$x[n] = 0$ for all $n < 0$.

anti causal = if it is defined for $n < 0$ or for both $n < 0$ & $n \geq 0$.
$x[n] \neq 0$. $n < 0$.

Basic continuous time signals.

① STEP SIGNAL.

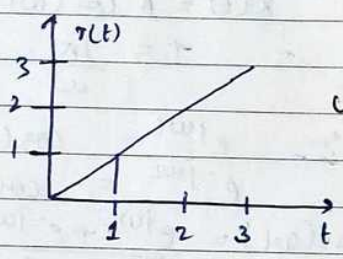
unit step signal:-
$u(t) = 1$ for $t \geq 0$.
$u(t) = 0$ for $t < 0$.



unit step signal.

② RAMP SIGNAL.

unit ramp signal
$r(t) = t$ for $t \geq 0$
$r(t) = 0$ for $t < 0$

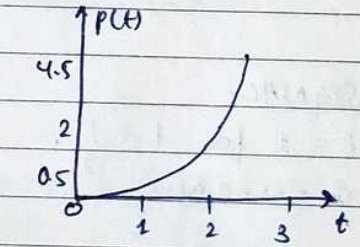


unit ramp signal.

③ PARABOLIC SIGNAL.

unit parabolic signal
$x(t) = \frac{t^2}{2}$ for $t \geq 0$.

$x(t) = 0$ for $t < 0$.



Relationship between these signal.

- unit ramp signal obtained by integrating unit step signal.

$$\int u(t)\,dt = \int 1\,dt$$
$$= t = r(t).$$

- parabolic integrated of Ramp signal.

$$p(t) = \int r(t)\,dt = \frac{t^2}{2} u(t).$$

Location: Date: / /

Step signal is derivative of Ramp.
$u(t) = \frac{d}{dt} r(t)$.

• Ramp is derivative of Parabolic
$r(t) = \frac{d}{dt} p(t)$.

⇒ COMPLEX EXPONENTIAL & SINUSOIDAL.

Complex exponential ⇒ $x(t) = e^{(a+j\omega)t}$.
$= e^{at} \cdot e^{j\omega t}$.
e^{at} controls growth or decay; $e^{j\omega t}$ → pure rotation

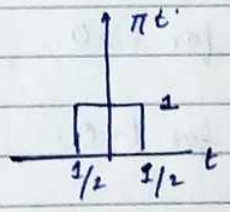
if $a=0$, complex sinusoidal.
$x(t) = e^{j\omega t}$.
$= e^{j\omega t} = \cos(\omega t) + j\sin(\omega t)$
$T = \frac{2\pi}{\omega}$ ⇒ pure complex exponential

if $a \neq 0$ → exponential growth/decay → non-periodic.

Sinusoidal Signals.
$x(t) = A\cos(\omega t + \phi)$ or $x(t) = A\sin(\omega t + \phi)$.
$T = \frac{2\pi}{\omega}$ → always periodic.

Relation :-
$e^{j\omega t} = \cos(\omega t) + j\sin(\omega t)$.
$e^{-j\omega t} = \cos\omega t - j\sin(\omega t)$.
$\cos(\omega t) = \frac{e^{j\omega t} + e^{-j\omega t}}{2}$ $\sin(\omega t) = \frac{e^{j\omega t} - e^{-j\omega t}}{2j}$.

UNIT PULSE SIGNAL.
$\pi t = 1$ for $t \leq \frac{1}{2}$.
$= 0$ elsewhere.



IMPULSE SIGNAL.
unit impulse signal.

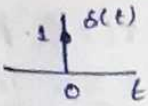
CONTINUOUS
$\delta(t) = 0$ for $t \neq 0$.
$\int_{-\infty}^{\infty} t\,dt = 1$.
$\delta(t) = \begin{cases} \infty & t=0 \\ 0 & t \neq 0. \end{cases}$

Sampling property :-
$\int_{-\infty}^{\infty} x(t)\,\delta(t-t_0)\,dt = x(t_0)$.

• property
$\int_{-\infty}^{\infty} x(t)\,\delta t\,dt = x(0)$.

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Location:



unit impulse.

Date: / /

DISCRETE.

$$\delta[n] = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}$$

Properties :-

① Shifting :- $\delta[n-n_0] = 1$ at $n = n_0$.

② Sampling :- $x[n]\delta[n-k] = x[k]$ at $n = k$.

Identity in convolution :-

$x[n] * \delta[n] = x[n]$.

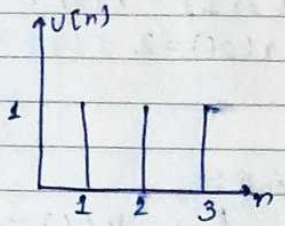
Basic Discrete Time signals.

Step signal.

unit step signal

$u[n] = 1$ for $n \geq 0$

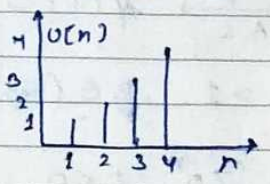
$= 0$ for $n < 0$.



Unit Ramp signal.

$r[n] = n$ for $n \geq 0$.

$= 0$ for $n < 0$.



Unit Impulse signal.

$\delta[n] = 1$ for $n = 0$.

$\delta[n] = 0$ for $n \neq 0$.

Transformation in independent variable of signals.

- Time scaling.
- time shifting.

- Time scaling :- compression or expansion of a signal in time.

$x(t) \xrightarrow{T.S} y(t) = x(\alpha t)$, $\alpha \neq 0$.

case- $|\alpha| > 1$ $\alpha \in (-\infty, -1) \cup (1, \infty)$. $\alpha \neq 0.1, 0.2$ ---

↳ Compression.

$x(t) \longrightarrow y(t) = x(\alpha t)$ ⇒ General eqn

Case - $|\alpha| < 1$ Expansion $\alpha \neq 0$. $\alpha \in (-1, 0) \cup (0, -1)$.

$\alpha = 0.1, 0.2 - - -$.

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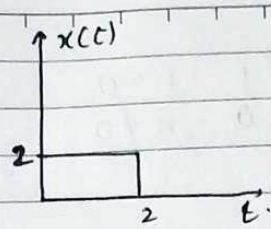
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$y(t) = x(\alpha t)$. eg: $y(t) = x(2t)$. peak at $t_0 = 3$.

$t_{new} = \frac{t_{old}}{a}$. $t_{new} = \frac{3}{2} = 1.5$.

Question:-

$x(t) = 2 \quad 0 \le t \le 2$.



$x = 2$.

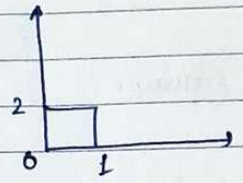
$t = 0$.

$x(2 \times 0) = x(0) = 2$.

$t = 1$

$x(2 \times 1) = x(2)$.

$x(2t) = 2$



$x(2t) = 2$.

$0 \le t \le 1$.

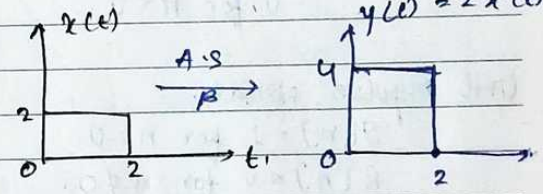
$\frac{0}{\alpha} \le \frac{2t}{\alpha} \le \frac{2}{\alpha}$.

Amplitude Scaling:-

$x(t) \xrightarrow[A.S]{\beta} y(t) = \beta x(t)$.

Case I = $|\beta| > 1$ $\beta \in (-\infty, -1) \cup (1, \infty)$. amplification.

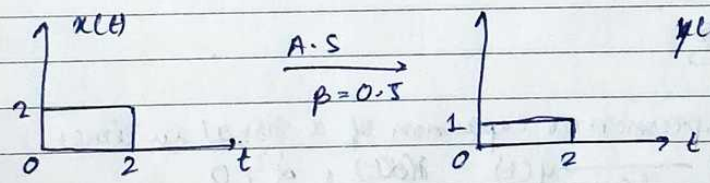
eg:- $x(t) = \begin{cases} 0 & t < 0 \\ 2 & 0 \le t \le 2 \\ 0 & t > 2 \end{cases}$.



$|\beta| = 2 > 1$.

Case II $|\beta| < 1$ $\beta \in (-1, 0) \cup (0, 1)$ $\beta = 0.1, 0.2 \ldots$

eg:-



$y(t) = 0.5x(t)$

* eg:-

$x(t) = \begin{cases} 2 & 0 \le t \le 3 \\ 0 & \text{Otherwise} \end{cases}$

amplitude scaling with 3.

$\rightarrow y(t) = 3x(t)$

$y(t) = \begin{cases} 3 \times 2 = 6 & 0 \le t \le 3 \\ 3 \times 0 & \text{otherwise} \end{cases}$

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Shifting :-

↳ Time

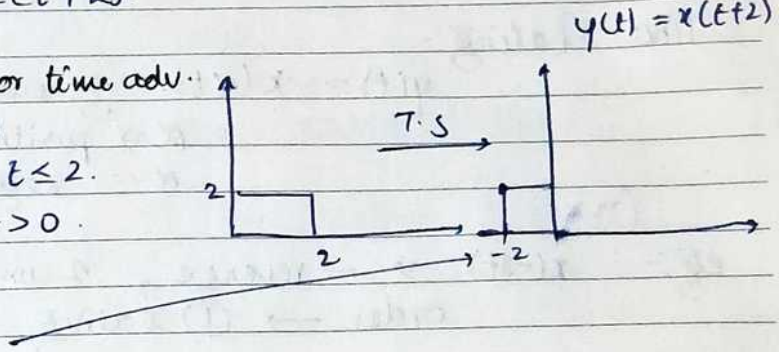
↳ Amplitude.

Time.

$x(t) \xrightarrow[k]{T.S} y(t) = x(t+k)$.

Case 1 → $k > 0$ left shifting or time adv.

$$x(t) = \begin{cases} 0 & t < 0 \\ 2 & 0 \le t \le 2. \\ 0 & t > 0. \end{cases}$$



$x(0) = 2 = y(-2)$.

$x(2) = 2. = y(0)$.

General form ⇒ $y(t) = x(t - t_0)$.

if :- $x(t - t_0)$ ⇒ Right shift by t_0.

if :- $x(t + t_0)$ ⇒ left shift by t_0.

Q :- $x(t) = u(t) - u(t-3)$ ← given find = $y(t) = x(t-4)$.

$u(t)$; $u(t-3)$.

$t - 3$ = right shift by 4.

$0 + 4 \longrightarrow 4$.

$3 + 4 \longrightarrow 7$.

$$x(t) = \begin{cases} 1 & 4 \le t \le 7. \\ 0 & \text{otherwise} \end{cases}$$

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Amplitude

General form ⇒ $y(t) = x(t) + c$.

$x(t) + c \longrightarrow$ upwards

$x(t) - c \longrightarrow$ downwards.

Time Scaling :-

$y(t) = x(\alpha t)$. General.

$\alpha \Rightarrow$ positive → no reverse.

$\alpha =$ negative → mirror image.

eg :- $x(-2t) \Rightarrow$ – reverse, 2 compression by 2.

order → ① scaling → ② reverse.

Time Shifting :-

Time shifting alone is never reversed.

only when the 't' has minus.

Reverse when :-

$x(-t)$, $(x)(-t+a)$, $x(-2t)$.

eg :- • $x(t+2) \longrightarrow$ no reverse

• $x(-t+4) \longrightarrow$ shift + reverse.

• $t-4 \longrightarrow$ right shift by 4. $-1(t-4) = -(t-4) \neq (t+4)$

• – reverse.

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Multiple transformation of cont. Time Signals.

General form = $Ax\left(\frac{-t-t_0}{a}\right)$.

Step 1 → $y(t) = Ax\left(\frac{-t}{a} - \frac{t_0}{a}\right)$.

Step 2 → plot $x(t)$. Given in prob.

Step 3 → plot $Ax(t)$. Amp scaling

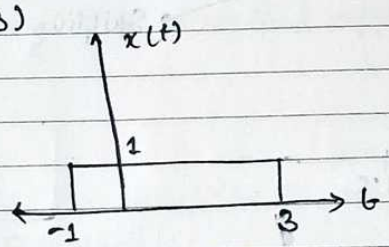
Step 4 → Plot $Ax(-t)$. Time reversal.

Step 5 → plot $Ax\left(-t-\frac{t_0}{a}\right)$. Time ~~scaling~~ shifting.

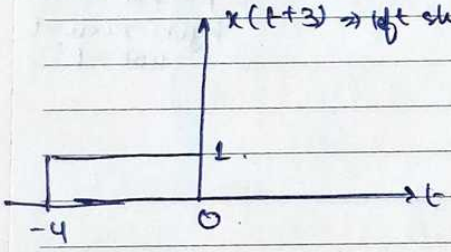
Step 6 → Plot $Ax\left(\frac{-t}{a} - \frac{t_0}{a}\right)$ Time scaling.

Eg: $y(t) = 3x(2t+3)$

$x(t) =$



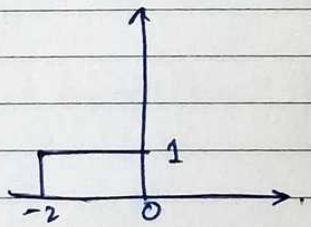
(1) $x(t) \xrightarrow[K=3]{\text{T. shift}} x(t+3)$.



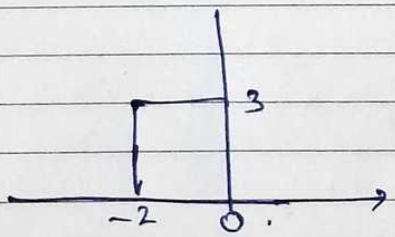
(2) $x(t+3) \xrightarrow[\alpha=2]{\text{scaling}} x(2t+3)$.

dividing -4 by 2 & 0 by 2.

$\frac{-4}{2} = -2 \quad \frac{0}{2} = 0$.



(3) $x(2t+3) \xrightarrow[\beta=3]{\text{Amp. scale}} 3x(2t+3)$.

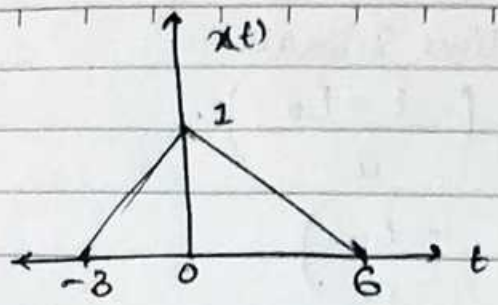


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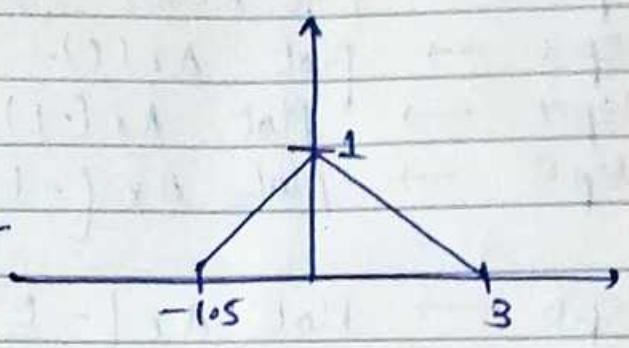
Eg: 2 $x(t) =$



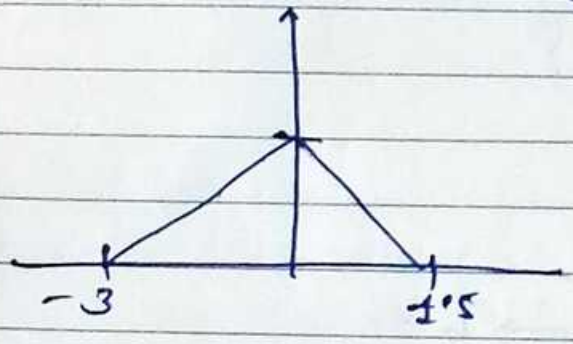
$y(t) = x(-2t+1)$.

- $x(t) \xrightarrow[\text{scaling}]{} x(2t)$.

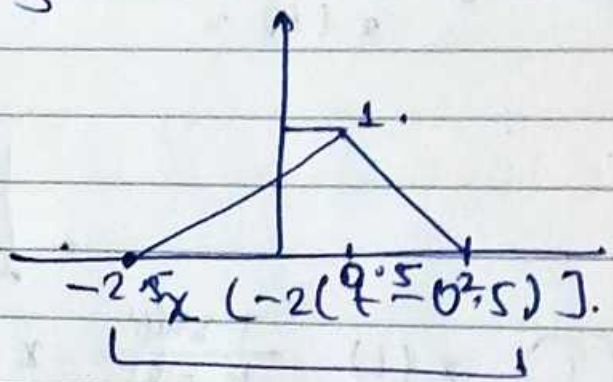
$-1.5, \frac{-3}{2}, \frac{6}{2} = 3$.



- Reversal.



Shifting. $x(-2t) \longrightarrow x(-2t+1)$



$x(-2(t-0.5))$.

t karna lai isko kyuki xaxis = t. not '-t'.

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FOURIER SERIES.

Condition for existence of Fourier series:-
exists only if the following Dirichlet's condition:-

- Signal $x(t)$ must be single valued function.
- signal must possess only a finite number of discontinuous in period T.
- Signal must have a finite num. of maxima & minima in period T.
- $x(t)$ must be absolutely integrable i.e $\int_0^T |x(t)|\,dt < \infty$.

Determination of Fourier Series ~~representation~~ representation of cont. time periodic signals:-

- CTFS.

any periodic continuous-Time signal can be expressed as a weighed sum of sinusoidal or complex exponentials.

(a) Trigonometric Fourier Series (Real form).

$$x(t) = a_0 + \sum_{k=1}^{\infty} \left(a_k \cos(k\omega_0 t) + b_k \sin(k\omega_0 t)\right)$$

Derivation:-

$$x(t) = a_0 + \sum_{k=1}^{\infty} \left(a_k \cos(\omega_0 t k) + b_k \sin(k\omega_0 t)\right)$$

$$\omega_0 = \frac{2\pi}{T}.$$

Derive for a_0.
integrate it from to to $t_0 + T$.

$$= \int_{t_0}^{t_0+T} x(t)\,dt = \int_{t_0}^{t_0+T} \left[a_0 + \sum_{k=1}^{\infty} \left(a_k \cos(k\omega_0 t) + b_k \sin(k\omega_0 t)\right)\right] dt.$$

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- $\int a_0 dt = a_0 T$.
- $\int_T \cos k\omega_0 t \, dt = 0$.
- $\int_T \sin k\omega_0 t \, dt = 0$.

because sin t cos have equal +ve & -ve area over a period.

$\therefore \int x(t) dt = a_0 T$.

So, $a_0 = \frac{1}{T} \int_{t_0}^{t_0+T} x(t) dt$.

Derive for a_k.

multiply both side by $\cos(m\omega_0 t)$:-

$$x(t)\cos(m\omega_0 t) = a_0 \cos(m\omega_0 t) + \sum_{k=1}^{\infty} [a_k \cos(k\omega_0 t)\cos(m\omega_0 t) + b_k \sin(k\omega_0 t) \cos(m\omega_0 t)]$$

Integrate over one period :-

$$\int x(t)\cos(m\omega_0 t) dt = a_0 \int \cos(m\omega_0 t) dt + \sum_{k=1}^{\infty} a_k \int \cos(k\omega_0 t)\cos(m\omega_0 t) dt + \sum_{k=1}^{\infty} b_k \int \sin(k\omega_0 t)\cos(m\omega_0 t) dt$$

Using orthogonality :

$$\int_T \cos(k\omega_0 t)\cos(m\omega_0 t) dt = \begin{cases} 0 & k \neq m \\ \frac{T}{2} & k = m \end{cases}$$

$$\int_T \sin k\omega_0 t \cos m\omega_0 t \, dt = 0.$$

also,

$$\int \cos(m\omega_0 t) dt = 0.$$

$k = m$.

$\therefore \int x(t)\cos(m\omega_0 t) dt = a_m \times \frac{T}{2}$

$$a_m = \frac{2}{T} \int_{t_0}^{t_0+T} x(t)\cos(m\omega_0 t) dt$$

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$m \to k$:

$$a_k = \frac{2}{T}\int_{t_0}^{t_0+T} x(t)\cos(k\omega_0 t)\,dt.$$

Derive for b_k :

both sides multiply by $\sin(m\omega_0 t)$:

$$x(t)\sin(\omega_m t) = a_0\sin(m\omega_0 t) + \sum_{k=1}^{\infty}\left[a_k\cos(k\omega_0 t)(\cos m\omega_0 t) + b_k \sin k\omega_0 t)\sin m\omega_0 t\right]$$

Using orthogonality :

$$\Rightarrow \int x(t)\sin(m\omega_0 t)\,dt = b_m \cdot \frac{T}{2}$$

$$\therefore b_m = \frac{2}{T}\int_{t_0}^{t_0+T} x(t)\sin(m\omega_0 t)\,dt.$$

$m \to k$,

$$b_k = \frac{2}{T}\int_{t_0}^{t_0+T} x(t)\sin(k\omega_0 t)\,dt.$$

Final eqn :

$$x(t) = a_0 + \sum_{k=1}^{\infty}(a_k\cos(k\omega_0 t) + b_k\sin(k\omega_0 t)).$$

$$a_0 = \frac{1}{T}\int_T x(t)\,dt. \qquad a_k = \frac{2}{T}\int_T x(t)\cos(k\omega_0 t)\,dt.$$

$$b_k = \frac{2}{T}\int_T x(t)\sin(k\omega_0 t)\,dt$$

⇒ Exponential form :

$$x(t) = \sum_{k=-\infty}^{\infty} C_k e^{jk\omega_0 t}.$$

C_k = complex Fourier coefficients.

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$$C_k = \frac{1}{T} \int_{t_0}^{t_0+T} x(t)\, e^{-jk\omega_0 t}\, dt.$$

⇒ Amplitude-phase form:-

$$x(t) = A_0 + \sum_{k=1}^{\infty} A_k \cos(k\omega_0 t + \phi_k).$$

where,

$$A_k = \sqrt{a_k^2 + b_k^2}.$$

$$\phi_k = -\tan^{-1}\left(\frac{b_k}{a_k}\right).$$

How to solve CTFS,

Step-01:- Find fundamental period T, → if sin wave, → LCM of periods

Step02:- compute $\omega_0 = \frac{2\pi}{T}$.

Step03:- Choose one period for integration

Piecewise → simplest piece

Symmetric → Symmetric limits $(-T/2$ to $T/2)$.

Step04:- plug into coefficient formula.

even signal:- only a_k non-zero.

Odd signal:- only b_k non-zero.

Mixed:- both

Step-05:- Write final series.

Standard signals CTFS:-

- Square wave:-

Only sine terms.

$$C_k = \frac{2}{jnK}\,(1-(-1)^K).$$

- Triangular wave.

Only cos

coefficient decay by $\frac{1}{k^2}$.

- Sawtooth wave.

Only sine.

- Exponential → $x(t) = e^{j\omega t}$.

$$C_k = \begin{cases} 1 & \text{if } k = \frac{\omega}{\omega_0} \\ 0 & \text{otherwise.} \end{cases}$$

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Properties of CTFS :-

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jw_0 kt} \qquad c_k e^{jw_0 kt}$$

① Linearity property :-

if $x_1(t) \longleftrightarrow C_{1k}$.

$x_2(t) \longleftrightarrow C_{2k}$.

Then

$$y(t) = a x_1(t) + b x_2(t).$$

Fourier coefficients are :-

$$Y_k = aC_{1k} + bC_{2k}.$$

Derivation :-

$$x_1(t) = \sum_{k=-\infty}^{\infty} C_k e^{jk\omega_0 t}$$

$$x_2(t) = \sum_{k=-\infty}^{\infty} C_{2k} e^{jk\omega_0 t}.$$

$$y(t) = a x_1(t) + b x_2(t).$$

put the values :-

after, $$y(t) = \sum_{k=-\infty}^{\infty} (aC_{1k} + bC_{2k}) e^{jw_0 kt}.$$

$\therefore$ $$y(t) = \sum_{k=-\infty}^{\infty} Y_k e^{jw_0 kt}.$$

$$\boxed{Y_k = aC_{1k} + bC_{2k}.} \quad *$$

② Time Shifting

if $y(t) = x(t - t_0)$.

$x(t) \longleftrightarrow C_k$.

Then $$Y_k = C_k e^{-jk\omega_0 t_0}. \quad *$$

Derivation :-

$$x(t) = \sum_{k=-\infty}^{\infty} C_k e^{jw_0 kt}.$$

$$y(t) = x(t - t_0).$$

Substitute $t - t_0$ everywhere,

$$y(t) = \sum_{k=-\infty}^{\infty} C_k e^{jw_0 k(t-t_0)}.$$

Location:

$$y(t) = \sum_{k=-\infty}^{\infty} C_k e^{jk\omega_0 t} e^{-jk\omega_0 t_0}$$

$$y(t) = \sum_{k=-\infty}^{\infty} \left(C_k e^{-jk\omega_0 t_0}\right) e^{jk\omega_0 t}.$$

$$y(t) = \sum Y_k e^{jk\omega_0 t}.$$

$$\boxed{Y_k = C_k e^{-jk\omega_0 t_0}}$$

③ Conjugate Symmetry:-

$$C_{-k} = C_k^* \text{ for all } k.$$

Derivation:-

$$x(t) = \sum_{k=-\infty}^{\infty} C_k e^{jk\omega_0 t}.$$

Take complex conjugate both sides.

Since $x(t)$ is real, $x(t)^* = x(t)$.

$$x(t)^* = \left(\sum_k C_k e^{jk\omega_0 t}\right)^* = \sum_k C_k^* (e^{jk\omega_0 t})^* = \sum_k C_k^* e^{-jk\omega_0 t}.$$

$x(t)^* = x(t)$ so,

$$x(t) = \sum_{k=-\infty}^{\infty} C_k^* e^{-jk\omega_0 t}.$$

put $m = -k$, so $k = -m$

$$x(t) = \sum_{m=-\infty}^{+\infty} C_{-m}^* e^{jm\omega_0 t}.$$

$$x(t) = \sum_{m=-\infty}^{\infty} C_m e^{jm\omega_0 t}.$$

$$C_m = C_{-m}^* \text{ for all } m.$$

or

$$\boxed{C_{-k} = C_k^*}$$

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④ Differentiation in time.

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t} \qquad \omega_0 = \frac{2\pi}{T}.$$

Define :- $y(t) = \frac{d}{dt} x(t)$.

Derivation :-

differentiate :-

$$y(t) = \frac{d}{dt}\left(\sum_k c_k e^{jk\omega_0 t}\right) = \sum_k c_k \frac{d}{dt} e^{jk\omega_0 t}$$

$$= \frac{d}{dt} e^{jk\omega_0 t} = jk\omega_0 e^{jk\omega_0 t}.$$

$$y(t) = \sum_k c_k (jk\omega_0) e^{jk\omega_0 t} = \sum_k (jk\omega_0 c_k) e^{jk\omega_0 t}.$$

$$\boxed{y_k = jk\omega_0 c_k} \quad *$$

⑤ Integration in time.

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}.$$

Define :- $y(t) = \int x(t)\, dt$.

Derivation :-

$$y(t) = \int \sum_k c_k e^{jk\omega_0 t} dt = \sum_k c_k \int e^{jk\omega_0 t} dt.$$

$$\int e^{jk\omega_0 t} dt = \begin{cases} \frac{1}{jk\omega_0} e^{jk\omega_0 t} & k \neq 0. \\ t & k = 0 \end{cases}$$

So,

$$\boxed{y(t) = c_0 t + \sum_{k\neq 0} c_k \frac{1}{jk\omega_0} e^{jk\omega_0 t} + K.} \quad *$$

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⑥ Parseval's Theorem:-

$$P = \frac{1}{T}\int_{t_0}^{t_0+T} |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |C_k|^2$$

Average power of time domain signal = sum of power in frequency domain harmonics.

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Discrete Time Fourier Series:-

$$x[n] = \sum_{k=0}^{N-1} C_k e^{j\frac{2\pi}{N}kn}$$

$$x[n] = \sum_{k=0}^{N-1} C_k e^{j\frac{2\pi}{N_0}kn}$$

where,

$$C_k = \frac{1}{N}\sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}$$

$$C_k = \frac{1}{N}\sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N_0}kn}$$

Eg:- Given sequence of period $N=4$.

$x[n] = \{1, 1, 0, 0\}$.

for $n = 0, 1, 2, 3$ determine DTFS C_k.

Ans:- period $= 4 = (N)$

$$\omega_0 = \frac{2\pi}{N} = \frac{2\pi}{4} = \frac{\pi}{2}.$$

Formula $= C_k = \frac{1}{N}\sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}$.

$$= \frac{1}{4}\sum_{n=0}^{3} x[n] e^{-j\frac{2\pi}{4}kn}.$$

$x[0] = 1 \quad x[1] = 1 \quad x[2] = 0 \quad x[3] = 0.$

When $n=0$,

$x[0] = 1.$

$e^{-j\frac{2\pi}{4}k(0)} = e^0 = 1.$

Term $= 1 \times 1 = 1.$

$n = 1.$ $\quad x[1] = 1.$

$e^{-j\frac{2\pi}{4}k(1)} = e^{-j\frac{\pi}{2}k}.$

Term $= 1 \times e^{-j\frac{\pi}{2}k}$

$n = 2 \Rightarrow 0 \Rightarrow$ Term

$n = 3 \Rightarrow 0 =$ term.

$$C_k = \frac{1}{4}\left(1 + e^{-j\frac{\pi}{2}k}\right)$$

Determination of DTFS.

Step-01:- Find Fundamental period N.

$$N = \frac{2\pi}{\omega_0}.$$

Step02:- Write one full period explicitly.

Step03:- Substitute $x[n]$ into DTFS.

$$C_k = \frac{1}{N}\sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}.$$

Step4 $\Rightarrow$ Simplify.

Step 5 $\Rightarrow$ Write final answer for all $k = 0 \ldots N-1$.

SHORTCUTS:-

- If sequence is real & even. Only cos $\rightarrow$ coeff real. $C_k = C_{-k}$
- Sequence is real & odd. Sine $\rightarrow$ coeff imag. $C_k = -C_k$
- Constant seq. $x[n] = A \Rightarrow C_0 = A$, $C_{k\neq 0} = 0$.

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$k=0$,

$$c_0 = \frac{1}{4}(1+1) = \frac{1}{2}.$$

$k=1$.

$$c_1 = \frac{1}{4}\left(1+e^{-j\frac{\pi}{2}}\right) = \frac{1}{4}(1-j).$$

$k=2$,

$$c_2 = \frac{1}{4}\left(1+e^{-j\pi}\right) = \frac{1}{4}(1-1) = 0$$

$k=3$,

$$c_3 = \frac{1}{4}\left(1+e^{-j\frac{3\pi}{2}}\right) = \frac{1}{4}(1+j).$$

$$c_k = \frac{1}{4}\left(1+e^{-j\frac{\pi}{2}k}\right).$$

Ans

properties of DTFS.

① Linearity :-

If $x_1[n] \longleftrightarrow c_{1k}$, $x_2[n] \longleftrightarrow c_{2k}$.

then,

$$y[n] = ax_1[n] + bx_2[n].$$

$$y_k = ac_{1k} + bc_{2k}.$$

Derivation:-

$$x_1[n] = \sum c_{1k} e^{j\frac{2\pi}{N}kn}$$

$$x_2[n] = \sum c_{1k} e^{j\frac{2\pi}{N}kn}.$$

$$y[n] = ax_1[n] + bx_2[n].$$

$$y[n] = a\left(\sum c_{1k} e^{j\frac{2\pi}{N}kn}\right) + b\left[\sum c_{2k} e^{j\frac{2\pi}{N}kn}\right].$$

$$y[n] = \sum (ac_{1k} + bc_{2k}) e^{j\frac{2\pi}{N}kn}.$$

$$y[n] = \sum y_k e^{j\frac{2\pi}{N}kn}.$$

$$y_k = ac_{1k} + bc_{2k}$$

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② Time shifting:-

If $y[n] = x[n-n_0]$.

Then $Y_k = C_k e^{-j\frac{2\pi}{N}kn_0}$.

Derivation:-

$$x[n] = \sum C_k e^{j\frac{2\pi}{N}kn}.$$

$$y[n] = x[n-n_0].$$

$$y[n] = \sum C_k e^{j\frac{2\pi}{N}k(n-n_0)}.$$

$$y[n] = \sum C_k e^{j\frac{2\pi}{N}kn} \cdot x\, e^{j\frac{2\pi}{N}k(-n_0)}.$$

$$y[n] = \sum Y_k e^{j\frac{2\pi}{N}kn}.$$

$$\boxed{Y_k = C_k e^{-j\frac{2\pi}{N}kn_0}.}$$

③ Conjugate Symmetry:-

If $x[n]$ is real then,

$$C_{N-k} = C_k^*.$$

Derivation:-

$$x[n] = \sum_{n=0}^{N-1} C_k e^{j\frac{2\pi}{N}kn}.$$

$$x[n]^* = \sum C_k^* e^{-j\frac{2\pi}{N}kn}.$$

As $x[n]$ = real then $x[n]^* = x[n]$.

$$x[n] = \sum C_{N-k}^* e^{j\frac{2\pi}{N}kn}$$

Compare

$$x[n] = \sum C_k e^{j\frac{2\pi}{N}kn}.$$

$$\boxed{C_{N-k} = C_k^*}$$

④ Periodicity of DTFS coefficients.

$$C_{k+N} = C_k.$$

Derivation:-

$$C_{k+N} = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}(k+N)n}.$$

$$e^{-j\frac{2\pi}{N}(k+N)n} = e^{-j\frac{2\pi}{N}kn} \times e^{-j2\pi n}.$$

$$e^{-j2\pi n} = 1.$$

So,

$$C_{k+N} = \frac{1}{N} \sum x[n] e^{-j\frac{2\pi}{N}kn} = C_k.$$

$$\boxed{C_{k+N} = C_k.}$$

⑤ Parseval Theorem:-

$$\frac{1}{N} \sum_{n=0}^{N-1} |x[n]|^2 = \sum_{k=0}^{N-1} |C_k|^2.$$

Derivation:-

$$x[n] = \sum C_k e^{j\frac{2\pi}{N}kn}.$$

$$x[n]^* = \sum C_m^* e^{-j\frac{2\pi}{N}mn}.$$

Multiply:-

$$|x[n]|^2 = \sum_k \sum_m C_k C_m^* e^{j\frac{2\pi}{N}(k-m)n}.$$

$$P = \frac{1}{N} \sum_{n=0}^{N-1} |x[n]|^2.$$

$$P = \frac{1}{N} \sum_k \sum_m C_k C_m^* \sum_{n=0}^{N-1} e^{j\frac{2\pi}{N}(k-m)n}.$$

Using orthogonality:- $\sum_{n=0}^{N-1} e^{j\frac{2\pi}{N}(k-m)n} = \begin{cases} N & k=m \\ 0 & k \neq m \end{cases}$

$k = m \Rightarrow P = \sum = |C_k|^2$

$$\boxed{\frac{1}{N} \sum |x[n]|^2 = \sum |C_k|^2}$$

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Representation of CT signals by its sample.

⇒ **Sampling:-**

The process of converting a CT signal into a discrete time signal by taking its value at uniform spaced time instants.

$$x[n] = x(nT).$$

T = sampling period. $\quad f_s = \frac{1}{T}$ sampling freq.

n = integer.

Models of Sampling:-

① **Ideal impulse sampling:-**

In this, the CT signal is multiplied with an impulse train, producing samples represented as weighted impulses located at integer multiple of sampling period.

$$x_s(t) = x(t) \sum_{n=-\infty}^{\infty} \delta(t - nT). \quad \text{(Impulse Train)}.$$

$$x_s(t) = x(t) \cdot p(t). \quad \text{(Sample signal)}.$$

$$\text{Expand} = x_s(t) = \sum_{n=-\infty}^{\infty} x(nT)\, \delta(t - nT).$$

② **Natural Sampling:-**

When $x(t)$ is passed through short duration pulses.

$$x_s(t) = x(t) \cdot s(t). \quad \longrightarrow \text{pulse train.}$$

③ **Frequency domain Representation of Sampling:-**

If $X(f)$ is CTFT of $x(t)$ then,

$$X_s(f) = \frac{1}{T} \sum_{K=-\infty}^{\infty} X(f - Kf_s).$$

- Sampling replicates the spectrum periodically in frequency domain.
- If replicas overlap → Aliasing.

Relation between $x(t)$ & $x[n]$.

$$x[n] = x(nT).$$

$$\boxed{x(t) = \sum_{n} x[n]\, \delta(t - nT) * h(t).}$$

$\longrightarrow$ reconstruction filter.

Changes when we move from CT to DT:

$x(t) \longrightarrow x[n] = x(nT)$.

$F(Hz) \longrightarrow \omega$ (rad/sample).

FT $\longrightarrow$ DTFT.

bandlimit = B Hz $\longrightarrow$ digital radian freq $\omega = 2\pi F/F_s$.

$$\omega = 2\pi \frac{F}{F_s}$$

⇒ SAMPLING THEOREM.

It states that a CT signal that is band-limited to a maximum freq fm can be perfectly reconstructed from its sample if it is sampled at a rate greater than or equal to twice its highest freq component.

$f_s \geq 2f_m$. This required rate is called the Nyquist Rate.

Derivation:-

sampled signal $y(t) = x(t) \cdot \delta(t)$ ——— (1).

$$\delta(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_s t + b_n \sin n\omega_s t) \quad \text{——— (2)}$$

where,

$$a_0 = \frac{1}{T_s} \int_{-\frac{T}{2}}^{T/2} \delta(t)\, dt = \frac{1}{T_s} \delta(0) = \frac{1}{T_s}.$$

$$a_n = \frac{2}{T_s} \int_{-\frac{T}{2}}^{T/2} \delta(t) \cos n\omega_s\, dt = \frac{2}{T_s} \delta(0) \cos n\omega_s 0 = \frac{2}{T}$$

$$b_n = \frac{2}{T_s} \int_{-T/2}^{T/2} \delta(t) \sin n\omega_s\, dt = \frac{2}{T_s} (0) \sin n\omega_s 0 = 0.$$

put these in eqn (2).

$$\delta(t) = \frac{1}{T_s} + \sum_{n=1}^{\infty} \left(\frac{2}{T_s} \cos n\omega_s t + 0 \right).$$

Substitute $\delta(t)$ in eq ①.

$$y(t) = x(t) \cdot \delta(t).$$

$$= x(t)\left[\frac{1}{T_s} + \sum_{n=1}^{\infty}\left(\frac{2}{T_s}\cos n\omega_s t\right)\right].$$

$$= \frac{1}{T_s}\left[x(t) + 2\sum_{n=1}^{\infty}(\cos n\omega_s t)\, x(t)\right]$$

$$y(t) = \frac{1}{T_s}\left[x(t) + 2\cos\omega_s t \cdot x(t) + 2\cos 2\omega_s t \cdot x(t) + 2\cos 3\omega_s t \cdot x(t) - - -\right]$$

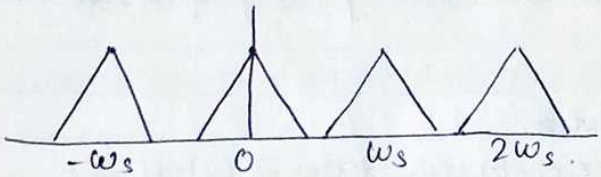
Take fourier Transform both side:-

$$Y(\omega) = \frac{1}{T_s}\left[X(\omega) + X(\omega-\omega_s) + X(\omega+\omega_s) + X(\omega-2\omega_s) - - -\right].$$

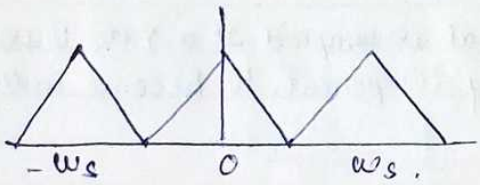
$$Y(\omega) = \frac{1}{T_s}\sum_{n=-\infty}^{\infty} X(\omega-\omega_s). \quad \text{where } n = 0, \pm 1, \pm 2 - - -$$

Cases:-

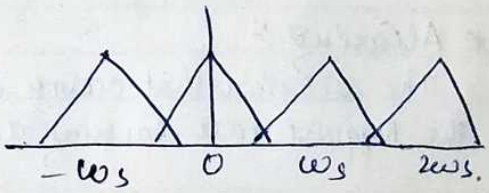
① $f_s > 2f_m$ oversampling.



② $f_s = 2f_m$ perfect sampling.



③ $f_s < 2f_m$ undersampling



Band limited meaning:-

A signal is band limited if:-

$$X(f) = 0. \text{ for } |f| > f_m.$$

meaning highest useful freq is f_m.

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Nyquist Rate & Nyquist Interval.

Minimum sampling freq required for perfect reconstruction.

$$f_N = 2f_m$$

Nyquist Interval :-

Sampling period at Nyquist rate :-

$$T_N = \frac{1}{2f_m}$$

Why is $f_s = 2f_m$ needed ?

If sampling is too small then → replicas overlap → aliasing.

If $f_s \geq 2f_m$.

Then replicas just touch but never overlap, so perfect reconstruction possible.

Condition for perfect reconstruction

$f_s \geq 2f_m$ & reconstruction filter is cutoff is :

$f_c = f_m$.

* Aliasing :-

The distortion that occurs when a signal is sampled at a rate lower than the Nyquist rate, causing different freq components to become indistinguishable

aliasing occurs due to :-

sampling creates spectral replicas :-

$$X_s(f) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X(f - k f_s).$$

If $f < 2f_m$

Then these replicas overlap & this overlap is called aliasing.

Condition for no aliasing :- $f_s \geq 2f_m$.

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Mathematical expression of aliasing:-

$$f_a = |f - kf_s|$$

f_a ← wrong freq; $|f - kf_s|$ → analog freq.

k some integer

* Eqn for:- if sampling freq is too low, f appears at f_a after sampling.

Reconstruction of signal from its sample:-

process of recovering the original CT signal $x(t)$ from its discrete sample $x[n] = x(nT)$ using an ideal low pass filter.

Sampling model:- $x_s(t) = \sum_{n=-\infty}^{\infty} x(nT)\,\delta(t - nT)$

frequency domain of sampling signal:-

$$X_s(f) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X(f - kf_s).$$

Reconstruction using an Ideal low pass filter:-

$H(f) = T$ for $|f| \le f_m$.
$H(f) = 0$ otherwise.
} reconstruction filter have freq response

$$Y(f) = H(f) X_s(f) = H(f) \frac{1}{T} \sum_{k=-\infty}^{\infty} X(f - kf_s).$$

$k = 0$ replicas lies inside $|f| \le f_m$.

$$Y(f) = X(f).$$

$$\therefore \quad y(t) = x(t).$$

* Time domain reconstruction formula.

Impulse response of ideal LPF is:-

$$h(f) = \text{sinc}\left(\frac{t}{T}\right).$$

Convolution with sampled impulse train

$$x(t) = \sum_{n=-\infty}^{\infty} x(nT)\, h(t - nT).$$

Substitute $h(t)$

$$x(t) = \sum_{n=-\infty}^{\infty} x(nT)\, \text{sinc}\left(\frac{t - nT}{T}\right).$$

} ideal reconstruction

Discrete time processing of CT signals.

Refers to the method in which a CT signal is first sampled then processed by a discrete time system and finally reconstructed back into a CT signal.

Sampling step ⇒ $x[n] = x(nT)$.

$f_s = \frac{1}{T}$

discrete time processing step.

impulse response $h[n]$.

Output sequence ⇒ $y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$.

In frequency domain (FTDT):-

$Y(e^{jw}) = X(e^{jw})\, H(e^{jw})$.

Reconstruction step :-

$$H_r(f) = \begin{cases} T, & |f| \le f_s/2 \\ 0 & \text{otherwise.} \end{cases}$$

Reconstructed signal :-

$$y(t) = \sum_{n=-\infty}^{\infty} y[n] \operatorname{sinc}\left(\frac{t-nT}{T}\right).$$

analog & Digital frequency relation :-

$$w = 2\pi \frac{f}{f_s} \quad \text{or} \quad f = \frac{w}{2\pi} f_s.$$

SAMPLING OF BANDPASS SIGNAL.

Process of sampling a bandpass signal at a rate lower than the Nyquist rate ($2f_m$) without causing aliasing.

$X(f) = 0$ for $f \notin [f_1, f_2]$.

Bandwidth = $f_2 - f_1$. Center frequency $f_c = \frac{f_1 + f_2}{2}$.

- bandpass sampling is possible because signal energy doesn't start at 0 Hz.

Bandpass sampling condition = $f_s \ge \frac{2B}{n}$.

* TYPES OF SYSTEMS.

① Linear & Non linear Systems.

When it satisfies superposition & homogenate principles.

- $T\{ax_1(t) + bx_2(t)\} = aT\{x_1(t)\} + bT\{x_1(t)\}$.

includes

Additivity = $T\{x_1 + x_2\} = T\{x_1\} + T\{x_2\}$ Homogeneity = $T\{ax\} = aT\{x\}$.

② Time Variant & Time Invariant.

If its input & output characteristics vary with time → TV.

Condition for variant ⇒ $y(n,t) \neq y(n-t)$

for Invariant = $y(n,t) = y(n-t)$.

③ Linear time variant & linear time invariant.

If a system is both linear & time variant ⟶ LTV.

If a system is both linear & time invariant ⟶ LTI.

④ Causality.

A system is causal if the output depends upon present & past inputs & doesn't depend upon future inputs.

⑤ Static & Dynamic

Static = memory-less. output depends upon present input.

dynamic = memory system. past value for the present input for which the system requires memory to get output.

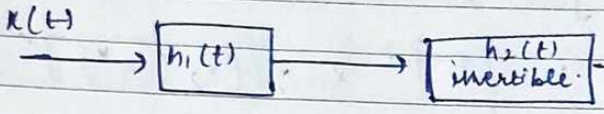
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Invertible & non invertible.

Invertible if the input of the system appears at the output.



Invertible system
$= x(t) = T^{-1}[y(t)]$.
$= T^{-1} T[x(t)]$.

$y(t) \neq x(t)$ non-vertible

Eg:- $Y(s) = X(s)\, H1(s)\, H_2(s)$.

$X(s)\, H1(s)\, \frac{1}{(H1(s)}$ Hence $H_2(s) = \frac{1}{H1(s)}$ Since.

$Y(s) = X(s)$

$y(t) = x(t)$. Invertible.

Stability

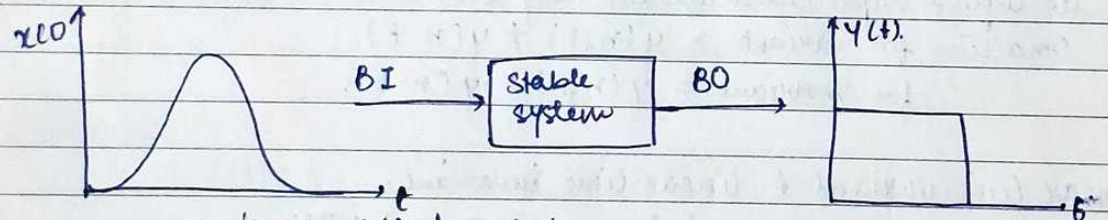
Stable :- when the output is bounded for bounded input.

For bounded input, if the output is unbounded in the system then it is said to be unstable.

For bounded signal, amplitude is finite.

Stable system or BIBO Stable System.

BIBO → BOUNDED INPUT BOUNDE OUTPUT.



input signal $x(t)$. bounded.

$|x(t)| < k_x < \infty$ for $-\infty < t < \infty$.

Where k_x is +ve real no.

$|y(t)| < k_y < \infty$.

FOR CT LTI,

$$\int_{-\infty}^{\infty} |h(t)|\, dt < \infty.$$

impulse response has finite area → System is stable

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LINEAR TIME INVARIANT SYSTEMS.

Continuous Time LTI

The input to the system is an impulse signal $x(t) = \delta(t)$.

Impulse response of the system is :-

$$y(t) = h(t) = T[\delta(t)]$$

$$x(t) = \int_{-\infty}^{\infty} x(\tau)\,\delta(t-\tau)\,d\tau.$$

$$y(t) = T[x(t)] = \int_{-\infty}^{\infty} x(\tau)\,T[\delta(t-\tau)]\,d\tau.$$

$$y(t) = T[x(t)] = \int_{-\infty}^{\infty} x(\tau)\,h(t-\tau)\,d\tau \quad \text{①}$$

Convolution Integral.

$$y(t) = x(t) * h(t).$$

for causal system limits will be $\int_0^t$

Discrete-Time LTI

$$x[n] = \delta[n].$$

$$y(n) = h(n) = T(\delta(n)).$$

$$x[n] = \sum_{k=-\infty}^{\infty} x(k)\,\delta(n-k)$$

$$y(n) = T[\delta(n)] = \sum_{k=-\infty}^{\infty} x(k)\,T(\delta(n-k)).$$

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)\,h(n-k)$$

Convolution Sum.

$$y(n) = x(n) * h(n).$$

also,

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)\,h(n-k) = \sum_{k=-\infty}^{\infty} x(n-k)\,h(k).$$

FREQUENCY RESPONSE OF LTI system.

- If input $= e^{j\omega t}$, output $= H(j\omega)\,e^{j\omega t}$.

$$H(j\omega) = H(s)\,|_{s=j\omega}.$$

($H(s)$: transfer function.)

$$H(j\omega) = |H(j\omega)|\,e^{j\angle H(j\omega)}.$$

Magnitude response $= |H(j\omega)|$
amplification / attenuation at ω.

Phase response $= \angle H(j\omega)$
- phase shift add.